

Notes: Mankiw & Zeldes (JFE, 1991)

The Consumption of Stockholders and Nonstockholders

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1 Big picture

- **Question.** Is it appropriate to test the consumption CAPM using *aggregate* consumption when most households do not own stock and therefore are not at the interior portfolio solution that generates the Euler equation?
- **Setting.** The paper combines household panel data from the **Panel Study of Income Dynamics (PSID)** with aggregate stock-return data. It uses food consumption from **1970 to 1984** and stockholding information from the **1984 PSID** wealth module.
- **Core conceptual problem.** The standard consumption CAPM uses the first-order condition for asset choice. But households that hold zero stock are at a corner, so their consumption should not enter the pricing relation in the same way as the consumption of stockholders.
- **Main contribution.** The paper is the first empirical attempt to split household consumption into **stockholders** and **nonstockholders** and ask whether the two groups have different covariation with the stock market.
- **Main descriptive fact.** In the 1984 PSID, only **27.6%** of families report positive stockholdings. Roughly **72.4%** own no stock at all.
- **Main empirical design.** The authors construct group-level per-capita food consumption for stockholders and nonstockholders, compute the correlation, volatility, and covariance of each group's consumption growth with the excess stock-market return, and compare those moments across groups under three stockholding thresholds.
- **Main result.** Stockholders' consumption is both **more volatile** and **more correlated with the excess return on equity** than nonstockholder consumption. Under the strictest stockholder definition, the covariance of consumption growth with the excess return is **0.002270** for stockholders versus **0.000305** for nonstockholders – more than seven times larger.
- **Statistical result.** In regressions testing equality of covariances, the slope coefficient is positive in all splits and is statistically significant for the stricter stockholder definitions: **0.079** with **p = 0.037** for the \$1,000 split and **0.090** with **p = 0.019** for the \$10,000 split.
- **Asset-pricing implication.** Using all PSID families implies a coefficient of relative risk aversion of about **100.4**. Using only stockholders in the strict split lowers the implied coefficient to about **35.2**. The paper therefore moves substantially toward explaining the equity premium, although it does not fully solve the puzzle.
- **What the paper is *not* doing.** It is not estimating a causal effect of stock ownership on consumption volatility. It is instead asking which households' consumption should matter for the Euler equation that prices equity.
- **Main limitations.** The data cover only **food consumption**, only **13 annual growth observations**, and classify households using **1984 stockholding status for the whole sample period**. These limitations likely attenuate the measured differences between groups.
- **Economic takeaway.** Representative-consumer asset-pricing tests can be badly misleading when market participation is incomplete. The consumption of **actual stockholders** appears much more relevant for pricing equities than the consumption of the population as a whole.

2 Data and measurement (who owns stock and how consumption is measured)

2.1 Why market participation matters for the Euler equation

- The standard consumption CAPM is derived from the first-order condition of an individual choosing a stock position optimally.
- If a household holds no stock, it is typically at a **corner solution**. Its marginal utility need not satisfy the Euler equation for equity holdings.
- Grossman and Shiller's aggregation result implies that one can aggregate consumption only across consumers who are at interior solutions with respect to the relevant asset.

Interpretation. The paper's key point is simple: if only stockholders are actually pricing stocks, then the relevant consumption series is *not* aggregate household consumption. It is the consumption of the subset of households that participates in the stock market.

2.2 PSID data on stockholdings and liquid assets

- The 1984 PSID survey is the first PSID wave to include detailed financial-wealth questions.
- One question asks for the market value of "shares of stock in publicly held corporations, mutual funds, or investment trusts, including stocks in IRAs."
- Another question asks about other liquid assets such as checking accounts, savings accounts, money-market funds, certificates of deposit, government savings bonds, and Treasury bills.
- The stockholding table is based on **2,998 families** in the 1984 survey.

Interpretation. The 1984 wealth module makes it possible to identify direct stock market participation and to compare it with broader liquid-asset holdings.

2.3 How common is stock ownership?

- **27.6%** of families hold a positive amount of stock; **72.4%** hold none.
- Only **23.2%** of families hold stock worth more than \$1,000.
- Only **11.9%** of families hold stock worth more than \$10,000.
- Families with no stock still account for about **62%** of disposable income, **68%** of food expenditure, and **34%** of total liquid assets.
- Among nonstockholders, **43.2%** have less than \$1,000 in other liquid assets. These households make up **31.3%** of the full sample.
- Yet lack of wealth is not the whole story: among households with more than \$100,000 in nonstock liquid assets, only **47.7%** hold any stock.

Interpretation. Many households avoid stocks because they are poor and possibly liquidity constrained, but a nontrivial share of relatively wealthy households also avoid stocks. That pattern is consistent with fixed participation costs, information costs, or noneconomic barriers to stockholding.

2.4 Who becomes a stockholder?

- Stockholding rises strongly with **income**. Across all education groups, the fraction of households owning stock is **11.1%** in the lowest labor-income quartile, **21.8%** in quartile 2, **31.6%** in quartile 3, and **52.2%** in the highest quartile.
- Stockholding also rises with **education**. For example, in income quartile 3, only **17.5%** of households with no high-school degree own stock, compared with **51.2%** of households with a college degree.
- Among households whose head has a high-school degree but no college degree, stock ownership rises from **14.1%** in the lowest labor-income quartile to **47.8%** in the highest.

Interpretation. The education and income gradients support the idea that stockholding involves fixed costs. Richer households have more to gain from paying them, and more educated households likely face lower information-processing costs.

2.5 Consumption measure and sample window

- The PSID asks about spending on **food consumed at home** and **food consumed in restaurants**. It does *not* ask about total consumption expenditures.
- The authors deflate each food component by the corresponding consumer price index and sum them to obtain real food consumption.
- The survey is conducted between late February and April, and the reported spending is treated as consumption during the **first quarter** of the year.
- Certain food-consumption questions were not asked in the first and sixth PSID surveys. As a result, growth rates are unavailable for the second, fifth, and sixth years.
- The final time series therefore contains **13 annual observations** on consumption growth between **1970 and 1984**.

Interpretation. The sample is short and noisy, but it is the first direct attempt to construct separate aggregate consumption series for stockholders and nonstockholders.

2.6 How stockholder and nonstockholder consumption are built

- Households are classified using their **1984 stockholding status**.
- The paper considers three splits:

1. **Split 1:** stockholder if stockholdings $> \$0$.
 2. **Split 2:** stockholder if stockholdings $\geq \$1,000$.
 3. **Split 3:** stockholder if stockholdings $\geq \$10,000$.
- For each year, family consumption is summed across stockholders, across nonstockholders, and across all families.
 - Each total is divided by the total number of family members in the corresponding group to form a **group-level per-capita consumption series**.
 - Excess stock return is measured as the return on the **S&P 500 minus the three-month Treasury bill rate**.

Interpretation. The paper is not tracking individual Euler equations. It is creating aggregate consumption series for groups that differ in their stock-market participation and then comparing their moments with equity returns.

2.7 Validation and descriptive consumption facts

- Stockholding families spend about **25% more per capita on food** than nonstockholding families.
- They spend about **12% more** on food at home and almost **80% more** on food away from home.
- About **25%** of stockholders' food spending and **17%** of nonstockholders' food spending occurs away from home.
- The PSID aggregates line up reasonably well with NIPA food data: the correlation is **0.61** for total food consumption, **0.75** for food at home, and **0.51** for food away from home.
- Average real food consumption per capita is \$1,323 in the PSID versus \$1,692 in the NIPA over the comparable period.

Interpretation. The PSID is noisy and narrower than the national accounts, but it is sufficiently aligned with aggregate food-consumption data to make the group comparison informative.

2.8 Measurement problems acknowledged by the paper

- Some households may hold stock indirectly through pension plans and still appear as nonstockholders.
- The authors argue this is not a major issue historically because in 1978 only **16%** of the labor force had defined-contribution pension plans.
- The much more serious issue is that stockholding status is measured only in **1984**. Because stock ownership rose sharply between 1965 and 1985, some households classified as stockholders in 1984 were likely nonstockholders earlier in the sample.

Interpretation. The likely direction of bias is **attenuation**. Misclassifying earlier nonstockholders as stockholders makes the two groups look more similar than they truly were.

3 Models and empirical specifications (all equations + interpretation)

3.1 Consumer optimization and the Euler equation

The representative stock-market participant chooses a portfolio to maximize

$$E_t \int_0^\infty e^{-\delta s} U(C_{t+s}) ds, \quad (1)$$

where C is consumption, $U(C)$ is instantaneous utility, and δ is the subjective rate of time preference. The first-order condition for asset i is

$$E_t \left[\frac{U'(C_{t+s})}{U'(C_t)} e^{-\delta s} (1 + R_{t,t+s}^i) \right] = 1. \quad (2)$$

Interpretation. This is the core pricing equation: assets that pay off more in states where marginal utility is high should command lower expected returns.

3.2 CRRA utility and the return-growth relation

The paper assumes isoelastic utility,

$$U(C) = \frac{C^{1-A}}{1-A}, \quad (3)$$

where A is the coefficient of relative risk aversion. A second-order approximation then implies

$$E(R_t^i) = AE(GC_t) + \delta - \frac{1}{2}A(A+1) \text{var}(GC_t) + A \text{cov}(R_t^i, GC_t), \quad (4)$$

where GC_t is the growth rate of consumption.

Interpretation. Expected returns are higher when an asset covaries more positively with consumption growth, because the asset then performs badly in bad times and is therefore riskier.

3.3 The cross-asset pricing relation

For any two assets i and j ,

$$E(R_t^i - R_t^j) = A \text{cov}(R_t^i - R_t^j, GC_t). \quad (5)$$

If asset i is the stock market and asset j is the risk-free bill, this becomes the equity-premium equation

$$E(R_t^m - R_t^f) = A \text{corr}(R_t^m - R_t^f, GC_t) \sigma(GC_t) \sigma(R_t^m - R_t^f). \quad (6)$$

Equivalently,

$$A = \frac{E(R_t^m - R_t^f)}{\text{cov}(GC_t, R_t^m - R_t^f)}. \quad (7)$$

Interpretation. The equity premium puzzle appears when the covariance between aggregate consumption growth and excess stock returns is too small. Then one needs an implausibly large A to rationalize the observed premium.

3.4 Why aggregate consumption may be the wrong object

- The usual empirical implementation uses **aggregate consumption of all households**.
- But nonstockholders do not satisfy the first-order condition for stockholdings, because they are not at an interior solution for the stock market.
- In a two-group economy with stockholders and nonstockholders, the relevant Euler equation should hold for the **total consumption of stockholders**, not for the total consumption of everybody.

Interpretation. The paper is not adding an ad hoc demographic split. It is redefining the empirical object so it matches the theory more closely.

3.5 Calibration with aggregate data

The paper first shows how severe the equity-premium puzzle looks under standard aggregate data:

- Using Mehra-Prescott annual data for 1890–1979 and nondurables plus services consumption implies $A \approx 26.3$.
- A time-aggregation correction lowers this to roughly **17.5**.
- Using postwar 1948–1988 nondurables plus services data raises the implied coefficient to about **89.0**.
- Using postwar food consumption gives **66.5**.
- Using aggregate PSID food consumption for all families gives **100.4**.

Interpretation. Before the stockholder split, the paper reproduces the standard message: the covariance of aggregate consumption with equity returns is far too small.

3.6 Empirical comparison of group-specific moments

For each stockholder split, the paper computes three moments for stockholders, nonstockholders, and the full sample:

1. $\text{corr}(GC_t, R_t^m - R_t^f)$,
2. $\sigma(GC_t)$,

3. $\text{cov}(GC_t, R_t^m - R_t^f)$.

Interpretation. The covariance is the economically decisive moment. It can rise because a group's consumption is more volatile, because it is more synchronized with the stock market, or both.

3.7 Regression test for equality of covariances

To test whether stockholder and nonstockholder covariances differ, the authors estimate

$$GC_t^S - GC_t^N = \alpha + \beta(R_t^m - R_t^f), \quad (8)$$

where GC_t^S is stockholder consumption growth and GC_t^N is nonstockholder consumption growth. The slope satisfies

$$\beta = \frac{\text{cov}(GC_t^S, R_t^m - R_t^f) - \text{cov}(GC_t^N, R_t^m - R_t^f)}{\text{var}(R_t^m - R_t^f)}. \quad (9)$$

Interpretation. $\beta = 0$ if and only if the two covariances are the same. A positive β means stockholder consumption comoves more with the excess stock return.

3.8 Economic magnitude of the covariance regression

- In split 2, the estimated slope is about **0.079**, which the authors interpret as follows.
- If the excess return on equity is **20%** – about one standard deviation from the mean – stockholder consumption growth rises by about **1.6 percentage points** relative to nonstockholder consumption growth.

Interpretation. The differences are not just statistically detectable. They are large enough to matter quantitatively for asset-pricing moments.

3.9 Why the paper uses the growth rate of the group average

- The baseline results use the growth rate of the **average consumption** of each group.
- An alternative would be to average household-level growth rates instead.
- That alternative gives qualitatively similar patterns but is much less statistically significant.

Interpretation. The preferred measure is the one most analogous to the aggregate consumption series used in the consumption-CAPM literature.

4 Identification and interpretation (what makes the evidence informative)

4.1 This is a pricing comparison, not a causal design

- The paper does not ask whether buying stock *causes* a household's consumption to become more volatile.
- The question is instead: *whose marginal utility should price stocks?*
- Because the theory points directly to market participants, the empirical design is a comparison of the relevant pricing group against the irrelevant aggregate comparison group.

Interpretation. The paper's logic is structural rather than causal. The test is informative if stockholders are closer to the marginal investors whose Euler equation prices equity.

4.2 Why the evidence likely understates the true difference

- Stockholding status is observed only in 1984, but the consumption sample begins in 1970.
- Stock market participation rose sharply over time, so some 1984 stockholders were probably nonstockholders earlier in the sample.
- This error mixes the two groups and pulls the estimated consumption moments toward each other.

Interpretation. A key strength of the paper is that its most obvious measurement problem biases against finding a difference. The positive results therefore carry more weight.

4.3 Why the short sample is a serious limitation

- The main consumption-growth sample has only **13 annual observations**.
- Food is only a slice of total consumption.
- Survey-based household consumption is noisy and subject to measurement error.

Interpretation. The results should be treated as suggestive rather than definitive. Small-sample noise makes it difficult to pin down asset-pricing moments precisely.

4.4 Why wealthy nonstockholders matter conceptually

- If all nonstockholders were poor and liquidity constrained, then the stockholder/nonstockholder split would be less informative about participation frictions among unconstrained households.
- But the paper documents many wealthy nonstockholders, including households with large liquid-asset balances who still avoid stock.

Interpretation. The presence of wealthy nonparticipants suggests that incomplete participation is not just a story about poverty. It is also a story about costs of entering or remaining in the stock market.

4.5 Why the paper only partially resolves the equity premium puzzle

- The stockholder-based covariance is much larger than the aggregate covariance.
- Even so, the implied coefficient of relative risk aversion remains around **35.2**, which is still too high to be comfortable.
- The paper therefore narrows the puzzle but does not eliminate it.

Interpretation. The paper's contribution is to show that incomplete participation is a quantitatively important piece of the puzzle, not that it is the entire answer.

4.6 Why the long-sample implication is only suggestive

- In the PSID sample, the covariance of stockholder consumption with excess returns is about three times the covariance for aggregate consumption.
- The authors note that if the same ratio held for the long 1890–1979 sample, a coefficient of relative risk aversion around **6** could explain the historical equity premium.
- But this is an extrapolation, not a direct estimate, because the long sample lacks panel data on stockholder consumption.

Interpretation. The long-run implication is intriguing and potentially important, but the paper is careful not to oversell it.

4.7 What better data would need to look like

- A better test would observe **total consumption**, not just food.
- It would observe **stockholding status repeatedly over time**, not only at the end of the sample.
- It would ideally cover a much **longer time series** to estimate covariances with far greater precision.

Interpretation. The paper is best seen as a proof of concept: once one conditions on market participation, the consumption data line up much more closely with asset-pricing theory.

5 Tables and figures

5.1 Table 1: Distribution of stockholdings and liquid assets

Read.

- The table is built from **2,998** PSID families in 1984.

- **72.4%** of households report zero stockholding; only **27.6%** report any stock at all.
- Only **11.9%** hold at least \$10,000 of stock.
- Many nonstockholders have almost no liquid wealth, but not all. Among households with more than \$100,000 in nonstock liquid assets, only **47.7%** own stock.

Economic message. Limited stock-market participation is quantitatively central. The pricing-kernel problem cannot be studied under the fiction that almost everyone is already a market participant.

Table 1
Distribution of stockholdings and liquid assets for U.S. households in 1984.
Based on 2,998 families in the 1984 Panel Study of Income Dynamics (PSID). Values are in 1984 dollars. The figure in each cell is the percentage of the population with the characteristics of that cell.

Stock value	Liquid assets (excluding stocks)					Total: Frequency #observations
	\$0	\$1–\$999	\$1,000–\$9,999	\$10,000–\$99,999	\$100,000 and up	
\$0	14.0%	17.3	25.0	14.6	1.5	72.4
\$1–\$999	0.2	0.9	2.5	0.7	0.1	4.5
\$1,000–\$9,999	0.4	1.0	5.1	4.3	0.4	13.4
\$10,000–\$99,999	0.1	0.5	2.8	6.1	0.5	11.3
\$100,000 and up	0.0	0.1	0.5	1.0	0.3	3.8
Total:						55
Frequency	14.7	19.8	35.9	26.8	2.9	100.0
# observations	440	592	1077	803	86	2998
% owning stock	4.8	12.3	30.6	45.5	47.7	27.6

Table 2
Fraction of U.S. families that hold stock in 1984 by education and average labor income.
Entries are the fraction of families in the corresponding cell that have stockholdings greater than zero. Numbers in parentheses are the total number of families (stockholders plus nonstockholders) in the corresponding cell. Labor income is equal to an average of the years 1981 to 1983. This table is based on families included in all of the 1982 to 1984 PSID surveys.

Quartiles of average labor income	Education				Total (# observations)
	No high school degree	No college degree	College degree	Advanced degree	
Quartile 1	2.7%	14.1	36.7	28.6	11.1
Quartile 2	11.2	20.2	42.5	42.3	21.8
Quartile 3	17.5	28.7	51.2	33.3	31.6
Quartile 4	48.7	47.8	60.8	50.8	52.2
Retired	12.9	30.5	58.8	48.1	25.7
Total	12.4	27.7	53.5	45.1	28.4
(# observations)	(591)	(1184)	(316)	(142)	(2233)

5.2 Table 2: Stockholding by education and labor income

Read.

- Stock ownership rises sharply with income: from **11.1%** in the lowest labor-income quartile to **52.2%** in the highest.
- It also rises with education. For example, in income quartile 3, stock ownership is **17.5%** for households without a high-school degree and **51.2%** for households with a college degree.
- Among households with a high-school degree but no college degree, ownership rises from **14.1%** in the lowest income quartile to **47.8%** in the highest.

Economic message. Participation costs look real: richer households and more educated households are much more likely to pay them.

5.3 Table 3: Calibrating the equity premium with aggregate data

Read.

- Using aggregate data on nondurables plus services from 1890–1979 gives an implied risk-aversion coefficient of **26.3**, or **17.5** after a time-aggregation correction.
- Using postwar 1948–1988 data raises the implied coefficient to **89.0** for nondurables plus services and **66.5** for food only.

- Using aggregate PSID food consumption for all families gives **100.4**.

Economic message. The standard representative-consumer implementation generates implausibly large risk aversion because aggregate consumption comoves too little with the stock market.

Table 3

Calibrating the equity premium: Aggregate data.

GC is the growth of consumption and A is the coefficient of relative risk aversion implied by the corresponding estimates.

	$\rho(GC, r^m - r^f)$	$\sigma(GC)$	$\sigma(r^m - r^f)$	$\text{cov}(GC, r^m - r^f)$	$E(r^m - r^f)$	Implied value of A
1890–1979 ^a (nondurables + services)	0.40	0.036	0.167	0.002349	0.062	26.3 17.5 ^c
1948–1988 ^b (nondurables + services)	0.45	0.014	0.140	0.000898	0.080	89.0
1948–1988 ^b (food only)	0.39	0.022	0.140	0.001201	0.080	66.5
PSID all families	0.26	0.021	0.148	0.000796	0.080 ^d	100.4

^aBased on annual data. All numbers are calculated from Mehra and Prescott (1985) data.

^bBased on 1st quarter to 1st quarter growth rates in National Income and Product Accounts and Ibbotson and Sinquefeld data. r^m is the return on the S&P 500, and r^f is the return on three-month Treasury bills. In each case, returns are calculated as the quarterly average of the monthly, twelve-month log return. Arithmetic (not log) returns are used to calculate the mean excess return. Further details are in the appendix in Mankiw and Zeldes (1990).

^cAdjusted for time aggregation.

^dUses value from 1948 to 1988.

Table 4

A comparison of the consumption of stockholders and nonstockholders.

GC is the growth of consumption (based on the PSID) and $r^m - r^f$ is the difference between the return on the S&P 500 and the return on three-month Treasury bills. Data are for 1970 to 1984.

	$\text{corr}(GC, r^m - r^f)$	$\sigma(GC)$	$\text{cov}(GC, r^m - r^f)$
<i>Total sample</i>			
	0.260	0.021	0.000796
<i>Split 1: Stockholders have stockholdings > \$0</i>			
Nonstockholders	0.093	0.020	0.000270
Stockholders	0.319	0.031	0.001440
<i>Split 2: Stockholders have stockholdings ≥ \$1000</i>			
Nonstockholders	0.047	0.020	0.000137
Stockholders	0.410	0.031	0.001855
<i>Split 3: Stockholders have stockholdings ≥ \$10,000</i>			
Nonstockholders	0.102	0.020	0.000305
Stockholders	0.488	0.032	0.002270

5.4 Table 4: Stockholder versus nonstockholder consumption moments

Read.

- For the full PSID sample, the correlation of consumption growth with the excess return is **0.260**, the standard deviation is **0.021**, and the covariance is **0.000796**.
- Under split 3, nonstockholders have correlation **0.102**, volatility **0.020**, and covariance **0.000305**.
- Under split 3, stockholders have correlation **0.488**, volatility **0.032**, and covariance **0.002270**.
- The stockholder covariance is therefore more than **seven times** the nonstockholder covariance under the strict split.

Economic message. Stockholder consumption is both more cyclical and more exposed to stock-market risk. That is exactly the pattern required to make equity returns look less anomalous.

5.5 Table 5: Test for equality of covariances

Read.

- The estimated slope β is **0.054** in split 1, **0.079** in split 2, and **0.090** in split 3.
- The one-sided p-values are **0.132**, **0.037**, and **0.019**, respectively.
- The evidence strengthens as the stockholder definition becomes stricter.

Economic message. The covariance gap is not just driven by a loose stockholder definition. It becomes sharper when the stockholder group is restricted to households with more meaningful equity exposure.

Table 5

Test for equality of covariances.

$$GC_{\text{stockholders}} - GC_{\text{nonstockholders}} = \alpha + \beta(r^m - r^f).$$

GC is the growth of consumption (based on the PSID) and $r^m - r^f$ is the difference between the return on the S&P 500 and the return on three-month Treasury bills. In split 1, a household is a stockholder if it holds any stock at all. In split 2, a household is a stockholder if it holds at least \$1,000 of stock. In split 3, a household is a stockholder if it holds at least \$10,000 of stock. In each case, the families that do not satisfy the criterion are considered nonstockholders. Data are for 1970 to 1984. Standard errors are in parentheses.

	Split 1	Split 2	Split 3
Constant (α)	0.0028 (0.0065)	0.0019 (0.0057)	0.0026 (0.0054)
$r^m - r^f$ (β)	0.054 (0.046)	0.079 (0.040)	0.090 (0.038)
$\bar{R}^2$	0.03	0.19	0.28
p -value (one-tailed test)	0.132	0.037	0.019

Table 6

Calibrating the equity premium: Stockholders vs nonstockholders.

GC is the growth of consumption (based on the PSID) and $r^m - r^f$ is the difference between the return on the S&P 500 and the return on three-month Treasury bills. A is the coefficient of relative risk aversion implied by the corresponding estimates. Data are for 1970 to 1984.

	$\rho(GC, r^m - r^f)$	$\sigma(GC)$	$\sigma(r^m - r^f)$	$\text{cov}(GC, r^m - r^f)$	$E(r^m - r^f)$	Implied value of A
PSID all families	0.26	0.021	0.148	0.000796	0.080 ^b	100.4
PSID ^a nonstockholders	0.10	0.020	0.148	0.000305	0.080 ^b	261.9
PSID ^a stockholders	0.49	0.032	0.148	0.002270	0.080 ^b	35.2

^aBased on split 3 (a household is a stockholder if it holds at least \$10,000 of stock and a nonstockholder otherwise).
^bUses value from 1948 to 1988.

5.6 Table 6: Calibrating the equity premium using stockholders versus nonstockholders

Read.

- Using all PSID families implies $A \approx \mathbf{100.4}$.
- Using nonstockholders alone implies an even more extreme value of about **261.9**.
- Using stockholders alone implies a much smaller, though still high, value of about **35.2**.

Economic message. The stockholder split goes a long way toward aligning the consumption data with the equity premium, but it does not finish the job.

6 Short summary

- The paper argues that aggregate consumption is the wrong object for testing the consumption CAPM when most households do not own stock.
- In the PSID, only about one-quarter of families hold any stock.
- Stockholders' consumption is more volatile and much more correlated with excess equity returns than nonstockholder consumption.
- The covariance that prices equity is therefore much larger for stockholders than for the full population.
- This lowers the implied coefficient of relative risk aversion dramatically, from about 100 to about 35 in the PSID sample.
- The result is economically important but not a full solution to the equity-premium puzzle because the data are short, noisy, and limited to food consumption.

7 Conclusion

7.1 Core conclusion

The consumption of stockholders and the consumption of nonstockholders are not the same object for asset pricing. Once the paper isolates the consumption of households that actually participate in the stock market, the consumption CAPM performs much better.

7.2 Economic intuition

The stock market should be priced by the households whose portfolios actually contain stock. Those households appear to experience consumption growth that is both more volatile and more tightly linked to stock-market fluctuations. Aggregate consumption dilutes that pricing signal by averaging in the consumption of nonparticipants.

7.3 Takeaway

This paper is an early and influential demonstration that **limited participation matters for asset pricing**. It does not eliminate the equity-premium puzzle, but it shows that part of the puzzle is an empirical measurement problem: the wrong consumers were being used in the Euler equation.